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A Navigational Guide to Statistical Methods

LS100 — Module 02 · Data Science & Statistics

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LS100 READING GUIDE

A Navigational Guide to Statistical Methods

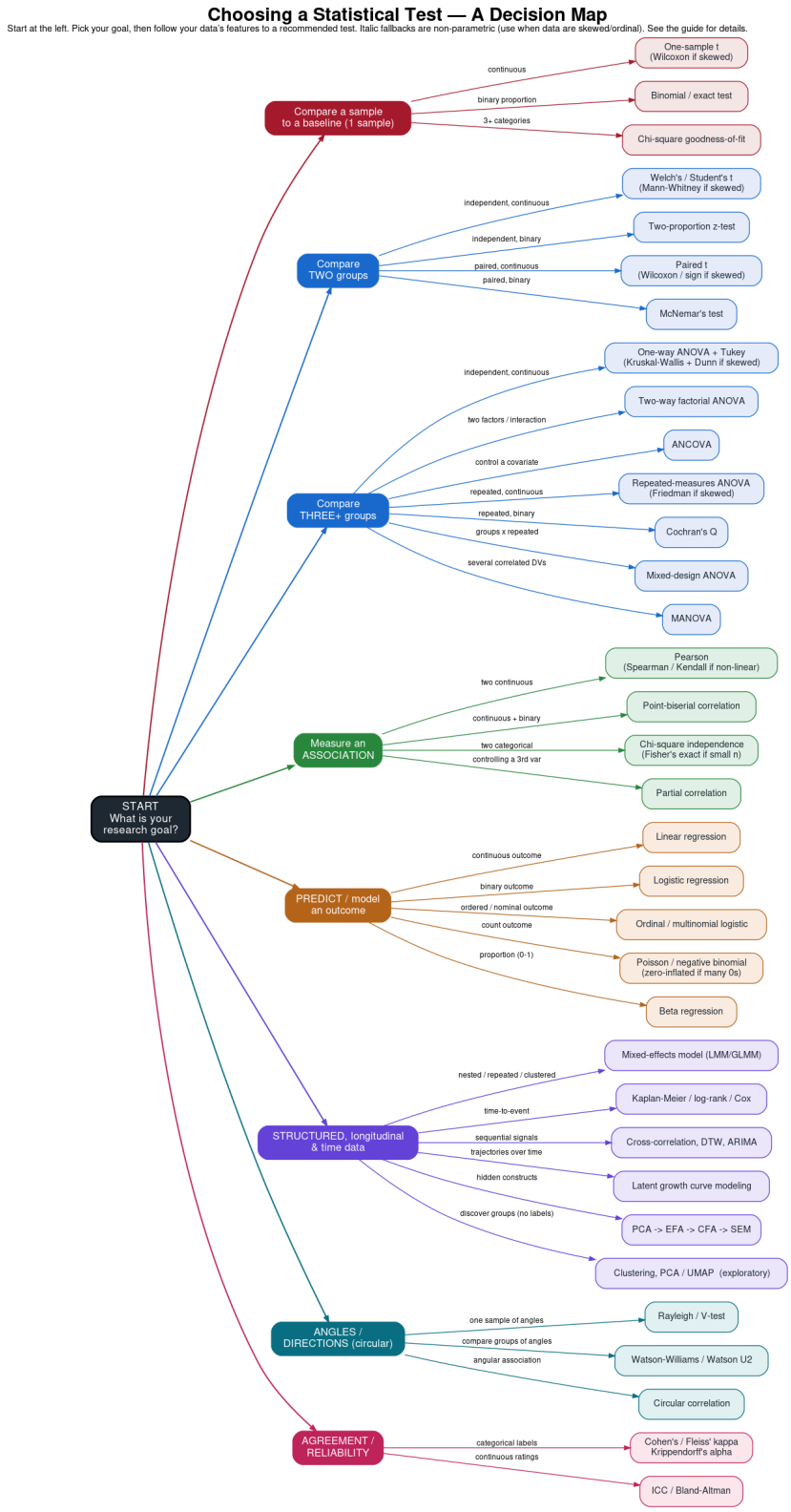
Souvik Mandal, Ph.D., Linkedin ID: souvik-mandal-phd

Project Leader & Instructor, Computational Behavioral Sciences, LS100, FAS, Harvard University

Behavioral data are often messy, count-heavy, zero-inflated, or tightly clustered within individuals, and rarely arrive in a clean, normally distributed package. Choosing the correct statistical tool requires diagnosing both the nature of your research question and the structural features of your data. This guide is a diagnostic map: instead of detailing the underlying mathematics, it tells you — given (a) your research question and (b) your data's features — (c) which statistical test family is appropriate. Within each objective, tests are ordered from the most general/basic to the most case-specific/advanced tests. Use it together with the one-page decision-tree poster that accompanies this guide.

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1. Step 0: Characterize Your Question and Your Data

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Before opening any test, answer the following five questions. Your answers point directly to a section below (and to a branch of the decision-tree poster).

1. **What is the goal?** compare a sample to a baseline (§1), compare groups (§2–§3), measure an association (§4), build a predictive model (§5), handle nested/longitudinal/time data (§6), analyze angles/directions (§7), or quantify rater agreement (§8).
2. **What type is the/ outcome (dependent) variable?** continuous, ordinal, nominal/binary, a count, a proportion, a time-to-event, or a circular angle.
3. **What is the predictor / grouping structure?** none (one sample), one grouping with two vs. three-or-more groups, one or more continuous predictors, or several factors at once.
4. **Are observations independent or related?** independent groups or paired/repeated/nested (the same individuals measured more than once).
5. **Is the outcome roughly normal, or skewed/ordinal?** this decides parametric vs. non-parametric — confirm it with the diagnostics in §9.

Please refer to *Table 01* for a quick glance of the appropriate statistical test for a given research question and features of the data.

Table 01: Quick-Reference for Appropriate Statistical Test

A fast lookup for the most common situations. Find your row, then read the detailed entry in the section indicated.

Your question / goal	Outcome type	Design	Go-to test(s) (§section)
Compare one mean to a fixed value	Continuous	One sample	One-sample t test, Wilcoxon if skewed (§1)
Compare one proportion to a value	Binary	One sample	Binomial / exact test (§1)
Category counts vs. expected	Nominal	One sample	Chi-square goodness-of-fit (§1)
Compare two group means	Continuous	2 independent	Welch's / Student's t-test, Mann-Whitney if skewed (§2)
Compare two paired means	Continuous	Paired	Paired t-test, Wilcoxon signed-rank if skewed (§2)
Compare two proportions	Binary	2 independent	Two-proportion z test, chi-square (§2)

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Your question / goal	Outcome type	Design	Go-to test(s) (§section)
Compare paired proportions	Binary	Paired	McNemar's test (§2, §4)
Compare 3+ group means	Continuous	3+ independent	One-way ANOVA + post-hoc test, Kruskal-Wallis if skewed (§3)
Compare 3+ repeated conditions	Continuous	Repeated	Repeated-measures ANOVA test, Friedman if skewed (§3)
Two factors + their interaction	Continuous	Factorial	Two-way ANOVA (§3)
Group means controlling a covariate	Continuous	1. covariate	ANCOVA (§3)
Several correlated outcomes at once	2+ continuous	Groups	MANOVA (§3)
Association of two continuous vars	Continuous	—	Pearson test, Spearman/Kendall if non-linear (§4)
Association of two categorical vars	Categorical	—	Chi-square test, Fisher's exact if small n (§4)
Predict a continuous outcome	Continuous	Predictors	Linear regression (§5)
Predict a binary / categorical outcome	Binary/nominal	Predictors	Logistic regression family (§5)
Predict a count	Count	Predictors	Poisson test, negative binomial if over-dispersed (§5)
Nested / repeated / clustered data	Any	Clustered	Mixed-effects model (LMM/GLMM) (§6)
Time-until-a-behavior	Duration + censoring	—	Kaplan-Meier / log-rank / Cox (§6)
Similarity of two movement sequences	Sequences	—	Cross-correlation, Dynamic Time Warping (§6)

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Your question / goal	Outcome type	Design	Go-to test(s) (§section)
Preferred direction / angle	Circular	—	Rayleigh test, Watson-Williams to compare (§7)
Agreement between coders / raters	Categorical or continuous	Raters	Cohen's/Fleiss' kappa test, ICC (§8)

2. Single-Sample Diagnostics: Comparing a Group to a Baseline

Use these when you have a single sample and want to know if it differs from a known baseline, a theoretical constant, or a chance probability.

1. **One-Sample t-test**

- **Alternative names:** Single-sample t-test
- **Data features:** One continuous variable; data should look roughly bell-shaped (normally distributed).
- **When to use:** To check if your sample's mean differs from a known or hypothesized population mean.
- **Behavioral example:** *Testing if the average response time of users on a new interface differs from the established industry baseline of 2.5 seconds.*

2. **One-Sample Wilcoxon Signed-Rank Test**

- **Alternative names:** Wilcoxon signed-rank test for a single sample
- **Data features:** One continuous or highly ordinal variable; heavily skewed or non-normal.
- **When to use:** The non-parametric alternative to the one-sample t-test, for data that violate normality or are ranked/ordinal.
- **Behavioral example:** *Testing if the median satisfaction score (1–7 Likert) after an intervention differs from a neutral score of 4.*

3. **Binomial / Exact Test**

- **Alternative names:** Exact binomial test
- **Data features:** One binary categorical variable (success/failure); valid at any sample size.
- **When to use:** To compare an observed proportion of a binary outcome against a hypothesized probability — the most basic single-sample categorical test.
- **Behavioral example:** *Testing whether an animal chooses the left arm of a T-maze more often than the 50% expected by chance across 20 trials.*

4. **Chi-Square Goodness-of-Fit Test**

- **Alternative names:** One-way chi-square test, multinomial goodness-of-fit test
- **Data features:** One categorical variable (nominal or ordinal) with three or more categories.

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- **When to use:** To determine if an observed frequency distribution across categories matches an expected/theoretical distribution.
- **Behavioral example:** *Investigating whether participants select three digital avatars with equal probability or prefer a specific one.*

3. Comparing Two Groups

These assess differences between exactly two groups, conditions, or time points.

1. **Independent (unrelated) groups**

    1. **Independent-samples t-test**

- **Alternative names:** Student's t-test, two-sample t-test, unpaired t-test
- **Data features:** One categorical IV (2 mutually exclusive groups); one continuous DV. Assumes normality and equal variances.
- **When to use:** The classic test to compare the average outcomes of two separate, independent groups.
- **Behavioral example:** *Comparing average test scores of an experimental learning group against a traditional control group.*

    2. **Welch's t-test**

- **Alternative names:** Welch's unequal-variances t-test
- **Data features:** One categorical IV (2 groups); one continuous DV. Does NOT assume equal variances or equal group sizes.
- **When to use:** The safer default for comparing two independent means — generally recommended over Student's t whenever variances or group sizes may differ.
- **Behavioral example:** *Comparing mean stride length between a small expert group and a larger novice group whose variability differs.*

    3. **Mann-Whitney U Test**

- **Alternative names:** Wilcoxon rank-sum test, Mann-Whitney-Wilcoxon (MWW) test
- **Data features:** One categorical IV (2 groups); one ordinal or non-normal continuous DV.
- **When to use:** The non-parametric equivalent of the independent-samples t-test, for skewed or ordinal outcomes.
- **Behavioral example:** *Comparing daily social-media posts between two groups, where counts are highly skewed (many users post 0).*

    4. **Two-Sample Z-test for Proportions**

- **Alternative names:** Two-proportion z-test
- **Data features:** One categorical IV (2 groups); one binary DV (e.g., Yes/No). Requires a large sample ($np \geq 10$ and $n(1-p) \geq 10$).
- **When to use:** Comparing the proportion of a specific outcome between two separate groups.

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- **Behavioral example:** *An A/B test comparing click-through rate between a blue and a green landing-page button.*

2. **Paired / related measurements**

    2. *3.a. Paired-Samples t-test*

- **Alternative names:** Dependent t-test, repeated-measures t-test, matched-pairs t-test
- **Data features:** Two continuous measurements from the same individuals (or matched pairs). Assumes the paired differences are normal.
- **When to use:** To evaluate the difference between two related means, as in pre-test/post-test designs.
- **Behavioral example:** *Measuring participant anxiety before and immediately after a stress-reduction session.*

3.a.i. *Sign Test:*

- **Alternative names:** Paired sign test
- **Data features:** Two related measurements; uses only the direction (not the magnitude) of each pair's difference. Minimal assumptions.
- **When to use:** The simplest paired test, for when you can only trust the sign of each change (ordinal or grossly non-normal differences).
- **Behavioral example:** *Whether posture symmetry improved (better/worse) after a training session, judged per participant.*

3.a.ii. *Wilcoxon Signed-Rank Test (Paired):*

- **Alternative names:** Wilcoxon matched-pairs signed-rank test
- **Data features:** Two related/paired measurements; ordinal or heavily skewed, non-normal differences.
- **When to use:** The non-parametric equivalent of the paired t-test, for pre/post or matched designs violating normality.
- **Behavioral example:** *Comparing self-reported urge to smoke (skewed 1–10 scale) before and after viewing a health-warning graphic.*

4. 3. Comparing Three or More Groups: Factorial & Multi-Variable Designs

When your design expands beyond two groups or includes multiple intersecting factors.

1. 5. Independent groups

    1. 6. One-Way ANOVA

- **Alternative names:** One-factor analysis of variance, single-factor ANOVA
- **Data features:** One categorical IV (≥ 3 groups); one continuous DV. Assumes normal residuals and equal variances.

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- **When to use:** To determine whether the means of three or more independent groups differ.
- **Behavioral example:** *Comparing attention-span scores across silent rooms, rooms with classical music, and rooms with open-office noise.*

6.a. Welch's ANOVA

- **Alternative names:** Heteroscedastic one-way ANOVA
- **Data features:** One categorical IV (≥ 3 groups); one continuous DV with unequal variances.
- **When to use:** A one-way comparison of 3+ means when variances differ across groups (the ANOVA analogue of Welch's t).
- **Behavioral example:** *Comparing mean call duration across three animal colonies whose variability differs markedly.*

6.b. Kruskal-Wallis Test

- **Alternative names:** Kruskal-Wallis one-way ANOVA by ranks, non-parametric one-way ANOVA
- **Data features:** One categorical IV (≥ 3 groups); one ordinal or non-normal continuous DV.
- **When to use:** The non-parametric alternative to one-way ANOVA, for skewed or ordinal outcomes.
- **Behavioral example:** *Comparing the ranking of product frustrations across three user-experience cohorts.*

6.c. Post-hoc Pairwise Comparisons

- **Alternative names:** Tukey HSD, Games-Howell, Dunn's test, Bonferroni/Holm correction
- **Data features:** Follows a significant omnibus test (ANOVA or Kruskal-Wallis).
- **When to use:** After a significant omnibus result, to find WHICH specific group pairs differ while controlling the family-wise error rate — Tukey HSD (equal variances), Games-Howell (unequal variances), Dunn's test (after Kruskal-Wallis).
- **Behavioral example:** *An ANOVA shows attention differs across three sound environments; Tukey HSD identifies which pairs actually differ.*

2. 7. Multiple factors and covariates

2. 8. Two-Way Factorial ANOVA

- **Alternative names:** Multi-way ANOVA, factorial analysis of variance
- **Data features:** Two or more categorical IVs (factors); one continuous DV.
- **When to use:** To examine the simultaneous effect of two factors and, critically, their interaction (whether one factor's effect depends on the other).
- **Behavioral example:** *Testing how reward type (cash vs. praise) and task difficulty (easy vs. hard) interact to affect performance time.*

8.a. ANCOVA

- **Alternative names:** Analysis of covariance

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- **Data features:** One or more categorical IVs; one continuous DV; one or more continuous covariates to control for.
- **When to use:** To compare group means while statistically controlling for a continuous confounding variable.
- **Behavioral example:** *Comparing post-test comprehension across three teaching methods while controlling for baseline pre-test scores.*

8.b. MANOVA

- **Alternative names:** Multivariate analysis of variance
- **Data features:** One or more categorical IVs; two or more related, correlated continuous DVs.
- **When to use:** To test whether groups differ on a combined multivariate profile, rather than running separate ANOVAs per outcome.
- **Behavioral example:** *Testing whether three personality types differ simultaneously across eye-fixation duration, click rate, and scrolling speed.*

3. 9. Repeated measures (same subjects)

3. 10. Repeated-Measures ANOVA

- **Alternative names:** Within-subjects ANOVA, dependent ANOVA
- **Data features:** The same subjects measured across ≥ 3 conditions or time points; one continuous DV. Assumes sphericity.
- **When to use:** When tracking change across multiple time points, or exposing the same participants to 3+ treatments.
- **Behavioral example:** *Tracking cognitive fatigue measured at hour 1, hour 3, and hour 5 of a continuous testing block.*

10.a. Friedman Test

- **Alternative names:** Friedman two-way ANOVA by ranks, non-parametric repeated-measures ANOVA
- **Data features:** One within-subjects factor (≥ 3 related conditions/time points); ordinal or non-normal continuous DV.
- **When to use:** The non-parametric alternative to repeated-measures ANOVA; pair with a Dunn's or Nemenyi post-hoc.
- **Behavioral example:** *Comparing a skewed comfort rating from the same users across three prosthetic settings.*

10.b. Cochran's Q Test

- **Alternative names:** Cochran Q
- **Data features:** One within-subjects factor (≥ 3 related conditions); one binary DV.
- **When to use:** The extension of McNemar's test to three or more repeated binary measurements.
- **Behavioral example:** *Whether the same animals succeed/fail at a task across three successive training days.*

10.c. Mixed-Design ANOVA

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- **Alternative names:** Split-plot ANOVA, between-within ANOVA, mixed factorial ANOVA
- **Data features:** At least one between-subjects factor (different groups) and at least one within-subjects factor (repeated measures); one continuous DV.
- **When to use:** The classic randomized-controlled-trial framework: does a treatment group change over time differently than a control group?
- **Behavioral example:** *Measuring depression scores at pre-test, post-test, and 6-month follow-up (within) for a treatment and a control group (between).*

11. 4. Associations, Relationships & Contingencies

These assess how variables change together, whether continuous scales or categorical labels.

11.0.i. *Pearson Correlation:*

- **Alternative names:** Pearson product-moment correlation coefficient (r)
- **Data features:** Two continuous variables. Assumes a linear relationship, normality, and homoscedasticity.
- **When to use:** To measure the strength and direction of a linear relationship between two continuous metrics.
- **Behavioral example:** *Relating total hours spent on an educational platform to final exam grade.*

11.0.ii. *Spearman's Rank Correlation:*

- **Alternative names:** Spearman's ρ (rho), rank-order correlation
- **Data features:** Two continuous or ordinal variables. No normality/linearity assumption; tracks monotonic relationships.
- **When to use:** The non-parametric alternative to Pearson, for curved relationships or ranked data.
- **Behavioral example:** *Correlating a user's self-reported stress rank with their rank of daily screen time.*

11.0.iii. *Kendall's Tau:*

- **Alternative names:** Kendall's τ coefficient, Kendall's rank correlation
- **Data features:** Two ordinal or continuous variables.
- **When to use:** A non-parametric alternative optimal for small samples or many tied ranks; more robust than Spearman in sparse data.
- **Behavioral example:** *Correlating preference rankings of 8 interface mockups between two small focus groups.*

11.0.iv. *Point-Biserial Correlation:*

- **Alternative names:** A special case of the Pearson correlation
- **Data features:** One continuous variable and one binary categorical variable.

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- **When to use:** To quantify the association between a continuous measure and a two-level grouping (equivalent to an independent t-test, expressed as a correlation).
- **Behavioral example:** *Relating movement smoothness to group membership (trained vs. untrained).*

11.0.v. *Partial Correlation:*

- **Data features:** Two continuous variables plus one or more continuous control variables.
- **When to use:** To measure the association between two variables while holding a third constant, removing its confounding influence.
- **Behavioral example:** *Relating practice hours to performance while controlling for baseline fitness.*

11.0.vi. *Chi-Square Test of Independence:*

- **Alternative names:** Pearson's chi-square test, chi-square bivariate test
- **Data features:** Two categorical variables. Large samples where every expected cell count is ≥ 5.
- **When to use:** To determine whether two categorical variables are significantly associated.
- **Behavioral example:** *Testing whether political affiliation is associated with preferred media medium (text, video, audio).*

11.0.vii. *Fisher's Exact Test:*

- **Alternative names:** Fisher-Irwin test
- **Data features:** Two categorical variables, typically a 2×2 table. Used when samples are small or events rare (any expected cell < 5).
- **When to use:** An exact calculation used when the large-sample approximation of the chi-square test breaks down.
- **Behavioral example:** *Whether a rare side effect is associated with a new drug vs. placebo in a small study.*

11.0.viii. *McNemar's Test:*

- **Alternative names:** McNemar's chi-square test
- **Data features:** Two paired/dependent nominal variables, structured as a 2×2 table of before/after binary states.
- **When to use:** To analyze changes in paired categorical proportions — also the standard test for comparing two classifiers on the same test set.
- **Behavioral example:** *Whether an ad campaign shifts voter preference (support/not) among the same individuals polled before and after.*

12. 5. Explanatory Modeling & Prediction: The Regression Family

When the goal shifts from testing differences to building predictive equations or isolating the contribution of multiple predictors. Most of these belong to the

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Generalized Linear Model (GLM) family, which differ mainly in the assumed distribution of the outcome.

12.0.i. *Simple Linear Regression:*

- **Alternative names:** Bivariate regression, straight-line regression
- **Data features:** One continuous predictor; one continuous outcome. Assumes linearity, independent and normal residuals, homoscedasticity.
- **When to use:** To predict a continuous outcome from a single predictor.
- **Behavioral example:** *Predicting annual charitable donation from a self-reported empathy score.*

12.0.ii. *Multiple Linear Regression:*

- **Alternative names:** Ordinary least squares (OLS) multiple regression
- **Data features:** Multiple continuous or dummy-coded predictors; one continuous outcome.
- **When to use:** To isolate the impact of several factors on a continuous outcome simultaneously, controlling for confounds.
- **Behavioral example:** *Modeling productivity from sleep, coffee, experience, and remote-work status.*

12.0.iii. *Binary Logistic Regression:*

- **Alternative names:** Logit regression, logistic modeling
- **Data features:** Multiple mixed predictors; one binary outcome (0/1).
- **When to use:** Predicting the probability of a binary event or choice.
- **Behavioral example:** *Modeling whether a shopper completes a purchase or abandons the cart from browsing time and discount.*

12.0.iv. *Ordinal Logistic Regression:*

- **Alternative names:** Ordered logit, proportional-odds model
- **Data features:** Multiple mixed predictors; one ordered categorical outcome (e.g., Low/Medium/High). Assumes proportional odds.
- **When to use:** Predicting an inherently ranked but non-continuous outcome.
- **Behavioral example:** *Predicting treatment compliance (non/partial/full) from demographic and psychological scales.*

12.0.v. *Multinomial Logistic Regression:*

- **Alternative names:** Polytomous logistic regression, multinomial logit
- **Data features:** Multiple mixed predictors; one nominal outcome with ≥ 3 unordered choices.
- **When to use:** Modeling discrete-choice behavior when options have no inherent ranking.
- **Behavioral example:** *Predicting which of four app themes a user selects from their personality and age.*

12.0.vi. *Poisson Regression:*

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- **Alternative names:** Log-linear count model
- **Data features:** Multiple mixed predictors; a count outcome (non-negative integers). Assumes the mean equals the variance.
- **When to use:** Modeling behavioral frequencies over a fixed window of time or space.
- **Behavioral example:** *Modeling monthly support tickets from software-utilization metrics.*

12.0.vii. *Negative Binomial Regression:*

- **Alternative names:** Over-dispersed count model
- **Data features:** Multiple mixed predictors; a count outcome whose variance exceeds its mean (over-dispersion).
- **When to use:** The standard correction when count data show heavy tails and extreme variation, violating Poisson assumptions.
- **Behavioral example:** *Modeling daily phone checks, where a few super-users inflate the variance.*

12.0.viii. *Zero-Inflated Models (ZIP / ZINB):*

- **Alternative names:** Zero-inflated Poisson, zero-inflated negative binomial
- **Data features:** Count outcomes with far more zeros than a standard distribution predicts.
- **When to use:** When zeros arise from two processes: 'structural zeros' (cannot/never perform the behavior) and 'true zeros' (could, but scored zero this time).
- **Behavioral example:** *Modeling daily cigarettes smoked: non-smokers always log 0, while quitting smokers may happen to log 0 on a given day.*

12.0.ix. *Beta Regression:*

- **Alternative names:** Proportional regression
- **Data features:** A continuous outcome strictly bounded between 0 and 1 (rates, proportions, fractions, excluding exactly 0 and 1).
- **When to use:** Predicting percentages or bounded allocations that show heteroscedasticity near the boundaries.
- **Behavioral example:** *Predicting the proportion of a monthly budget allocated to entertainment from psychometric scores.*

13. 6. Advanced: Clustered, Longitudinal, Survival & Time-Series Frameworks

For data that violate the assumption of independent observations, track hidden constructs, or unfold over time.

13.0.i. *Linear / Generalized Mixed-Effects Models (LMM / GLMM):*

- **Alternative names:** Hierarchical linear modeling (HLM), multilevel modeling (MLM), random-effects models

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- **Data features:** Data that are nested or clustered (repeated trials within participants; students within schools).
- **When to use:** Essential for behavioral data with dependencies. Splits effects into fixed effects (variables of interest) and random effects (baseline variation across individuals/clusters).
- **Behavioral example:** *Modeling how a learning game affects performance across 30 trials, allowing each player their own baseline skill.*

2. *13.a. Time-to-event (survival)*

    2. *13.b. Kaplan-Meier Estimator & Log-Rank Test*

- **Alternative names:** Kaplan-Meier survival curve; log-rank (Mantel-Cox) test
- **Data features:** A time-to-event outcome with right-censoring; one categorical grouping.
- **When to use:** To estimate 'time-until-behavior' curves and compare them between groups — the descriptive/comparative step before Cox modeling.
- **Behavioral example:** *Comparing how long until animals in two enrichment conditions first use a novel object.*

13.b.i. *Cox Proportional-Hazards Model:*

- **Alternative names:** Cox regression, survival analysis
- **Data features:** Predictors mapped to a time-to-event outcome with right-censoring.
- **When to use:** Modeling how predictors affect the time until a behavior or milestone occurs.
- **Behavioral example:** *Modeling days until a user cancels a subscription from engagement telemetry.*

3. *13.c. Sequential & time-series data*

    3. *13.d. Cross-Correlation Function*

- **Alternative names:** Lagged cross-correlation
- **Data features:** Two time-aligned continuous signals.
- **When to use:** To quantify how strongly two signals co-vary at different time lags — used to detect synchrony and lead-lag relationships between movement or physiological signals.
- **Behavioral example:** *Measuring whether one dancer's motion leads or follows a partner's, from the lag of peak correlation.*

13.d.i. *Dynamic Time Warping (DTW):*

- **Alternative names:** Elastic sequence matching (a similarity measure, not a hypothesis test)
- **Data features:** Two sequences (e.g., pose or gesture time series) that may differ in speed or timing.
- **When to use:** To measure similarity between two movement sequences while allowing local stretching/compression in time — foundational for gesture matching and behavior clustering.

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- **Behavioral example:** *Quantifying how closely a learner's tennis-serve trajectory matches an expert template despite a different tempo.*

13.d.ii. *ARIMA:*

- **Alternative names:** Box-Jenkins models; VAR for multiple parallel series
- **Data features:** High-frequency sequential observations from one entity, where values correlate with their own past (autocorrelation).
- **When to use:** Analyzing sensor streams or behavioral trends over continuous timelines to forecast future behavior.
- **Behavioral example:** *Forecasting next month's daily active users from three years of history, filtering weekly and seasonal cycles.*

4. 14. Latent structure & unsupervised discovery

    4. *14.a. Latent Growth Curve Modeling*

- **Alternative names:** Growth curve analysis, LGC analysis
- **Data features:** A structural-equation framework on longitudinal data with repeated measurements over explicit time frames.
- **When to use:** To model individual trajectories of growth, decline, or learning over time — not just group averages.
- **Behavioral example:** *Tracking how a child's socio-emotional development trajectory shifts year-over-year from ages 5 to 12.*

14.a.i. *Latent Structure Discovery (PCA arrow.r EFA arrow.r CFA arrow.r SEM):*

- **Data features:** Many correlated variables believed to reflect a smaller set of hidden constructs.
- **When to use:** A progression from data reduction to theory testing: PCA (compress correlated variables into uncorrelated components), EFA (discover hidden constructs), CFA (verify a predefined measurement structure), SEM (test complex multi-path theories with direct, indirect, and mediated effects).
- **Behavioral example:** *Reducing a battery of behavioral items to a few factors, then testing a theorized pathway among them with SEM.*

14.a.ii. *Unsupervised Pattern Discovery (Clustering & Dimensionality Reduction):*

- **Alternative names:** k-means, hierarchical clustering, silhouette score, PCA / UMAP / t-SNE (exploratory, not hypothesis tests)
- **Data features:** Many continuous features per observation, with no outcome labels.
- **When to use:** To DISCOVER groupings or compress structure rather than test a hypothesis — clustering groups similar observations, the silhouette score judges cluster quality, and PCA/UMAP/t-SNE reduce dimensions for visualization. Confirm any discovered groups with independent hypothesis tests.
- **Behavioral example:** *Clustering extracted audio embeddings to reveal vocalization types, or grouping postures from pose features.*

15. 7. Directional & Circular Data

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Angles and directions — joint angles, movement heading, time of day — live on a circle, where 0° and 360° are identical. Ordinary linear statistics mishandle this wrap-around (the mean of 350° and 10° is 0°, not 180°), so circular-specific methods are required. These are directly relevant to pose and movement analysis.

15.0.i. *Rayleigh Test:*

- **Data features:** One set of angles.
- **When to use:** To test whether angles are uniformly spread around the circle versus clustered around a single mean direction.
- **Behavioral example:** *Testing whether an animal's movement headings cluster in a preferred direction rather than being random.*

15.0.ii. *V-Test:*

- **Alternative names:** Modified Rayleigh test
- **Data features:** One set of angles plus a hypothesized direction.
- **When to use:** Like the Rayleigh test, but more powerful when you have an a-priori expected direction.
- **Behavioral example:** *Testing whether escape headings concentrate toward a known shelter direction.*

15.0.iii. *Watson-Williams Test:*

- **Alternative names:** Circular one-way ANOVA
- **Data features:** Angles from two or more groups, reasonably concentrated (von Mises distributed).
- **When to use:** To compare the MEAN directions of two or more groups — the circular analogue of ANOVA.
- **Behavioral example:** *Comparing mean limb-swing angle at toe-off between sprinters and joggers.*

15.0.iv. *Watson's U^2 Test:*

- **Alternative names:** Watson two-sample U^2 test
- **Data features:** Two sets of angles; non-parametric.
- **When to use:** To compare two circular distributions without assuming a von Mises shape.
- **Behavioral example:** *Comparing the distribution of head-orientation angles between two behavioral states.*

15.0.v. *Circular Correlation:*

- **Alternative names:** Circular-circular and circular-linear correlation
- **Data features:** Two angular variables, or one angular and one linear variable.
- **When to use:** To measure association involving angular data.
- **Behavioral example:** *Relating joint-angle phase to a continuous speed measure during gait.*

16. 8. Measurement Reliability & Agreement

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When several coders annotate behavior, or a measure is repeated, you must quantify how consistent those measurements are before trusting them — a routine step whenever a team builds an annotation schema.

16.0.i. *Cohen's Kappa:*

- **Data features:** Two raters; categorical (nominal) labels.
- **When to use:** To measure agreement between two coders beyond what chance alone would produce.
- **Behavioral example:** *Two students independently labeling each video frame as groom / walk / rest.*

16.0.ii. *Fleiss' Kappa:*

- **Data features:** Three or more raters; categorical labels.
- **When to use:** The extension of Cohen's kappa to more than two raters.
- **Behavioral example:** *A team of four annotators coding courtship behaviors.*

16.0.iii. *Krippendorff's Alpha:*

- **Data features:** Any number of raters; nominal, ordinal, interval, or ratio data; tolerates missing ratings.
- **When to use:** The most general agreement coefficient — useful when coders skip items or data types are mixed.
- **Behavioral example:** *Assessing agreement on an ordinal aggression-intensity scale where some clips were not coded by everyone.*

16.0.iv. *Intraclass Correlation Coefficient (ICC):*

- **Data features:** Continuous measurements; two or more raters, or repeated measurements.
- **When to use:** To quantify agreement/consistency for continuous ratings, and for test-retest reliability of a measure.
- **Behavioral example:** *Agreement among three raters' continuous limb-angle estimates, or the stability of an automated metric across repeated recordings.*

16.0.v. *Bland-Altman Analysis:*

- **Alternative names:** Limits-of-agreement plot
- **Data features:** Two continuous measurement methods applied to the same units.
- **When to use:** To visualize and quantify agreement between two measurement methods (e.g., manual vs. automated) — distinct from their correlation.
- **Behavioral example:** *Comparing joint angles from manual digitization against pose-estimation output.*

17. 9. Before You Test: Diagnostics, Robust & Resampling Methods

Choosing between parametric and non-parametric tests — and trusting any result — depends on first checking assumptions. These tools are how you decide.

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17.0.i. *Normality Checks:*

- **Alternative names:** Shapiro-Wilk test, Kolmogorov-Smirnov / Anderson-Darling, Q-Q plot
- **Data features:** One continuous variable, or a model's residuals.
- **When to use:** To judge whether the normality assumption holds. A small p-value (or a curved Q-Q plot) signals non-normality — consider a non-parametric or robust alternative.
- **Behavioral example:** *Checking whether stride-length residuals are normal before performing an ANOVA.*

17.0.ii. *Homogeneity-of-Variance Checks:*

- **Alternative names:** Levene's test, Bartlett's test, Brown-Forsythe
- **Data features:** A continuous variable across two or more groups.
- **When to use:** To test whether groups share equal variances (an assumption of Student's t and standard ANOVA). If violated, switch to Welch's t / Welch's ANOVA.
- **Behavioral example:** *Confirming equal variance of reaction times across conditions before a standard ANOVA.*

17.0.iii. *Permutation (Randomization) Tests:*

- **Alternative names:** Exact or Monte-Carlo permutation tests
- **Data features:** Almost any design.
- **When to use:** To obtain a p-value by repeatedly shuffling labels rather than assuming a theoretical distribution — ideal for small or oddly-shaped behavioral samples.
- **Behavioral example:** *Testing a group difference in a small movement study by permuting condition labels 10,000 times.*

17.0.iv. *Bootstrap:*

- **Alternative names:** Resampling with replacement
- **Data features:** Almost any statistic.
- **When to use:** To estimate confidence intervals and standard errors for any metric (means, medians, correlations, custom indices) without distributional assumptions.
- **Behavioral example:** *A 95% confidence interval for a custom gait-symmetry index by resampling participants.*

Always report an effect size (Cohen's d, η^2, r, odds ratio) alongside every p-value, and **correct for multiple comparisons** (Bonferroni, Holm, or FDR) whenever you run many tests.

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